Qualitative assessment of business processes

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Abstract There are various possibilities to evaluate the cost and time factors of processes. But these possibilities do not permit a quality assessment of business processes. For such processes, we need methods and performance measurement systems, which are not oriented solely on concrete measureable results. However, it is important to be able to assess and improve process quality.

The aim of the method is to determine the quality of a process area. In addition, areas of action are discovered, in which processes can and should improve, in order to reach a more mature level.

This method opens up a possibility for an organization to advance a continuous improvement of business processes with an assessment method to account for long term lasting effects. Furthermore, metrics will be established in order to measure and monitor the quality of processes and to provide a foundation which consists of a self-critical view on one's own processes.

Keywords
Business Process Management, Process Quality, Continuous Improvement

1 Introduction

On the way to lean and efficient business processes, the continuous improvement processes (CIP) gains importance. Since the actual achievement of an optimum is only possible in theory, the gradual improvement of the production is a good method to get the best possible result to approach this optimum. The challenge is to identify spheres of activities in order to improve and to derive appropriate action. The problem of detecting potential CIP is shown in the consideration of indirect processes. For the production of upstream processes, the production

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planning process (PPP) [1], it is inevitable to indeed make statements about the effectiveness, for example, by considering the transit time of the process or the use of labor expended. Statements about the effectiveness of achieving the result, and also about the quality of this process, are sometimes difficult and are not uniformly measurable. The aim of this work is an introduction of a method that will allow such a qualitative assessment of business processes.

2 Maturity Level

This model uses a six-level [2] structure for process evaluation, as used in the SPICE-standard. With this level structure, it is possible to define and measure the performance or the maturity level of an observed process area. A total of seven quality indicators were defined and four of them are described within this paper. For each of these indicators exists an individual interpretation of the content level. The business process can be subdivided into level 1 to 6, level 1 being the lowest level and level 6 the optimum aim image.

If a process is assigned during the appraisal of level 1, no process implementation will be possible and the process aim will not be achieved. Accordingly processes with the classification in level 6 run smoothly after a defined standard process. A continuous improvement of the process takes place. Illustration 1 offers an overview of the definition of the levels.

<u>Level 1: Incomplete</u>

The business processes are not performed correctly. Thereby there is no accurate calculation of metrics possible.

<u>Level 2: Partial</u>

The business process can be partially carried out or the figures can be elevated in some process steps. The process execution can’t be guaranteed.

<u>Level 3: Organized</u>

The implementation of the business process or the determination of these indicators is planned and reviewed.

<u>Level 4: Controlled</u>

A standard process is defined, which includes the tasks, inputs, outputs and the criteria to achieve the process goals.

<u>Level 5: Ascertainable</u>

It is estimated in advance, how changes in one process affect the process performance or the metrics. The process or the metrics run within defined limits.

Level 6: Optimal

There is a continuous improvement of the standard process. Furthermore the efficiency and the effectiveness of the business process will increase.

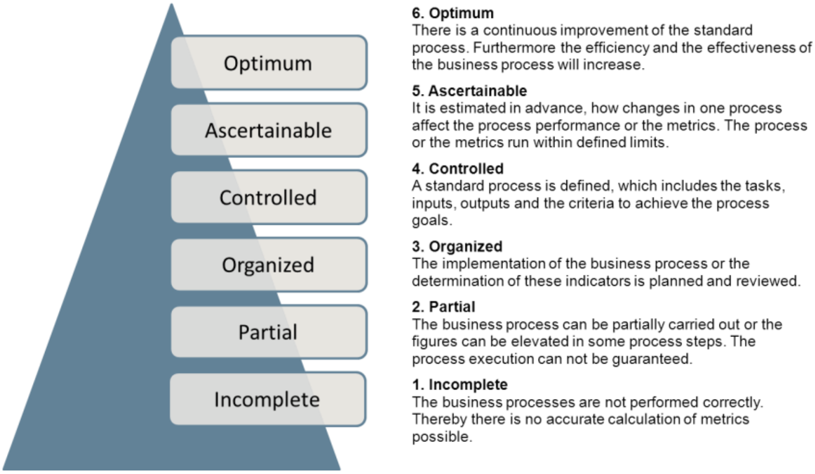

Fig. 1. Overview of the definition of the levels.

The prerequisite in order to be classified into a level is that all the criteria of the current level, as well as all the underlying levels, must be fulfilled. Indicators of the disclosure of possible improvements in the respective consideration area offer the criteria of the next higher level. Regular use of the assessment on comparable processes results in a development or improvement, because areas of action are identified and are considered as the progress of the processes under observation. It should be noted that a survey of key data can only be sustainably realized from the fourth level. This methodology provides the user with the possibility to raise metrics besides the degree of ripeness regulation.

3 Quality Indicators

Besides the maturity levels, six key performance indicators have been defined. For these indicators, there have also been defined individual interpretations of the maturity levels as well as the possibility for their improvement.

Currently the levels are used, to evaluate the quality of processes and the methodology in determining the level of maturity. Indicators can only be effectively investigated from level 4. These indicators can then be used for a continuous monitoring of the processes. By determining the levels of maturity, it is possible to see how the process is evolving. If indicators are raised additionally the process maturity can be supervised and there is the possibility to determine measures. Firstly, the definition of six indicators takes place in order to describe or investigate the quality of processes in the production planning.

3.1 Information by Documents

The identification number “Information by documents” expresses a high-quality of the information content of the passed on and required documents or working-step-results. Not only the adherence to delivery dates of the working-step-results is monitored, but also whether these are produced in a standardized form and how high the complaint rate of working-step-results is controlled. The maturity level structure is described as follows:

<u>Level 1: Incomplete</u>
There are no standard documents and templates for creating documents available. Important information which is needed for the realization of a process step is not transmitted by the process owner to the process partners. Consequently the achievement of the process result is not guaranteed.

<u>Level 2: Partial</u>
The documents and information flow conform to the minimum requirements to be able to fulfill the process purpose. There are some standard templates available where the required working-step-results are stored. Information is delivered to the executive process steps and the required information is transmitted at least on demand to the process partners.

<u>Level 3: Organized</u>
The requirements for the information content of required work results and deadlines are defined in advance. The backup of information content appropriate for requirement takes place through the principal use of standard templates or check lists which seize a completeness of the handed over work items.

<u>Level 4: Controlled</u>
The totality of the necessary working step results with completion date is known before start of the PPP. Therefore the part of standardized passing on of working step results and the adherence to schedules of delivered results can be determined. The documents are checked for the information content, i.e. the quality of the information capacity must be guaranteed in a way that process partners can carry out a process step without further problems. Information and documents reach to the intended place without detours.

<u>Level 5: Ascertainable</u>
The quality of the information content of working step results to be exchanged is guaranteed by the universal use of standard documents. The registration of any

working step result takes place once by which the relevance of present information is unequivocal. (The detection result of each step is done only once, so the relevance of existing information is clear).

Level 6: Optimal

In order to avoid media breaks and long waiting periods for needed information and documents, a system is provided, in which all process partners are involved. These can access the required documents during the process implementation and therefore information is made available.

3.1.1 Performance indicator for the quality indicator "Information by documents"

With the help of this value the adherence to delivery dates of required working step results can be controlled and supervised. A value less than 100% states that the set dates are not kept for the delivery by working step results. Therefore either new appointments must be defined or the processing time of the generation of working step results must be shortened. The identification number can be automatically raised, while the totality of the required working step results and the data comparison by date of delivery and plan appointment are taken from a project monitoring tool.

The formula for calculation is as follows:

$$Ita = \frac{AD}{AB}$$

With the proviso:

AD = Quantity as necessarily defined working step results with appointment term

AB = Quantity of the faithfully to appointment delivered required defined working step results

3.2 Degree of communication

The "degree of communication" expresses whether the communication is target-oriented in between process partners. Communication is a key to success. Without sufficient communication the project success cannot be guaranteed. All employees involved in the PPP must be informed of progress as well as important data about changes promptly and adequately.

Level 1: Incomplete

There is no project-oriented communication between the process owners and the process partners.

Level 2: Partial

A communication takes place now and then between the process partners. The responsibilities within the process are nominally known by the main process partners.

Level 3: Organized

The process owners are well known to the project participants. Regulated communication takes place in the form of project meetings if the process progress is endangered. Therefore contents and aims for these project meetings are known in general.

Level 4: Controlled

The process owners are known with their representatives. A regulated and regular communication takes place between all partners involved. In order to project meetings, participants functions are selected straight and invited as well as attendance lists are led. Further the contents and the aims of the meetings are defined in advance.

Level 5: Ascertainable

Periodical meetings take place in which the progress of the process is communicated and discussed. Besides, it is exactly determined at which time which process persons meet. The universal leadership of attendance lists allows the examination of the presence of necessary process partners. Comparisons of the achieved aims after the project meetings with the in advance planned ones take place.

Level 6: Optimal

An unequivocal process map exists for the communication which defines media, people, and appointments.

3.2.1 Performance indicator for the quality indicator "Degree of communication"

Herewith a monitoring of the aim reaching based on the communication during the control meetings takes place. It is checked whether the results from the control meetings fulfill to the goal which is set during a control communication. Trigger for a low value of the index can be, for example, that the set of goals are pinned up, so the goals should be broken down into sub-goals, in order to sharpen the measure. This can effectively be supported by a list of open tasks or PDCA (Plan-Do-Check-Act).

The formula for calculation is as follows:

$$Kze = \frac{AE}{AER}$$

With the proviso:
AE = Quantity of expected results from control meetings
AER = Quantity of achieved results from control meetings

3.3 Degree of process description

The “degree of the process description” describes how a process implementation proceeds on the basis of the available process description. It is to be supposed that with a holistic and detailed description of processes, a qualitatively high-quality result must be achieved.

<u>Level 1: Incomplete</u>
The processes within the PPP follow no predefined pattern. Descriptions that represent the processes and structures for all these process relevant information do not exist. Too many partial processes are involved in the process implementation to ensure the fulfillment of the process purpose.

<u>Level 2: Partial</u>
Process descriptions do exist fragmentarily. The success of the implementation of the whole process cannot be guaranteed by lacking knowledge about the detailed and networked expiry.

<u>Level 3: Organized</u>
Process descriptions exist which consider the complete process course and which are specified enough to ensure the success of the process implementation. The process is planned, supervised and explained conformal so that the work results of the process can be controlled and maintained.

<u>Level 4: Controlled</u>
The process descriptions are constructed at the risk of a defined action. The determination of process aims, process owners, form of application, process start and process goal, as well as identification numbers, which are needed to the examination of the process aim, takes place.

<u>Level 5: Ascertainable</u>

Defined processes are used on the basis of standard processes. For the defined metrics, aim values for the examination of the process result exist which admit a suitable valuation of the process success as well as initiating suitable measures in regard to misdemeanor.

<u>Level 6: Optimal</u>

The defined standards of the description of processes are continuously improved and applied generally. Metrics for verification of the target process are collected and monitored for each sub-process.

Action instructions exist for initiating the standard measures which have to be applied in case of any misdemeanor by process aims.

3.3.1 Performance indicator for the quality indicator "Degree of process description"

The inquiry of the identification number "degree of process description" is based on a realization of allocation matrix (see figure 2). It is to be chosen between the following scenarios:

1. **standard-process used**
2. **process description completely**
3. **process partially described**
4. **no description available**

Besides, every process can run "successful", "partially successful" or "not successful" (see figure 2 horizontal columns).

Line sheet to monitor the success in the execution of the subprocesses using the process description:

process	successful	partially successful	not succesful
standard process used	~~IIII~~ II		
description completely	IIII	III	
process partially descr.	III	II	I
No descr. available		II	III
Sum	14	7	4
Factors(defined)	1	0,5	0
Score	14	3,5	0

Σ 25 processes

$$\frac{\text{Score}}{\text{\# process running}} = \frac{17,5}{25} = 70\,\%$$

Fig. 2. Possibility for the evaluation of the indicator "degree of process description"

The formula for the calculation is:

$$Degree\ of\ pd = \frac{1 * successful + \frac{1}{2} * partially\ successful}{\sum process\ running}$$

3.4 Process stability

The Process stability expresses if process results can be predicted.

Level 1: Incomplete
The process is not able to deliver unchanged constant and predictable results on every process run.

Level 2: Partial
Unchanged and predictable results are delivered partly consistently, however, no reproduction of the process chain is ensured.

Level 3: Organized
The process delivers constant and predictable results of independent process partners.

Level 4: Controlled
It is possible to achieve punctual, reproducible, and traceable results which correspond to the process purpose.

Level 5: Ascertainable
Borders for the process stability are defined. Process stability may shift between these borders during the process implementation.

Level 6: Optimal
By a constant and consistent monitoring of the identification numbers of the process stability, the process delivers predictable and invariable results at any time. To guarantee this, defined measures are initiated, escalated and monitored having regard to their effects.

3.4.1 Performance indicator for the quality indicator "process stability"

To calculate the index "process stability" a tally sheet is to be used for the translation, because an automatic evaluation is currently not realizable. A comparison between the results of the carried out process steps and the previously defined result (in the process description) occurs. The line list exists of four scenarios (see figure 3) which are:

1. **exceeded**
2. **achieved within limits**
3. **achieved**
4. **missed**

The identification number can achieve an aim value of more than 100%, referred to the scenario "excelled". The possibility of an aim value of more than 100% of this metric has as both, positive and negative character. On the one hand excelling of 100% can be laid out as a bad planning and on the other hand as a high-class increase. The problems of this indicator lie with subjective consideration points of single process editors. If these are not capable to watch self-critically over their approach, a distorted result of the indicator is preprogrammed.

For the weighting of the result an allocation of defined factors is occurred. Figure 3 shows one way for the conversion of a line list.

	List to the predictability of the process results			
	exceeded	achieved	achieved within limits	missed
Geplantes Prozess-ergebnis wurde ...	𝍷𝍷𝍷	𝍸 𝍸 𝍸	𝍸 𝍷𝍷𝍷𝍷	𝍷𝍷𝍷
Sum	3	15	9	3
Factors (defined)	1,5	1	0,5	0
Score	4,5	15	4,5	0

$$\frac{\text{Score}}{\#\ \text{process results}} = \frac{24}{30} = 80\ \%$$

Fig. 3. Possibility for the evaluation of the indicator "degree of process description"

The formula for the calculation is:

$$stability = \frac{\frac{3}{2} * exceeded + 1 * achieved + \frac{1}{2} * achieved\ within\ limits}{\sum process\ results}$$

4 Expiry of the assessment

The duration of the assessment amounts in about one working day depending on how large the area of the assessed business process spans. One should make sure in advance that the area being evaluated does not fail because being too complex. Otherwise the results are not based on facts. If it is decided to carry out such an assessment it needs to be clarified why it is assessed and what processes are to be assessed. In addition, the involved persons are to be determined. The method of assessment of the PPP requires a continuous participation of individual employees, departments of planning, a CIP-coach, or employee of a master planned area, and usually a union representative. A non-continuous participation is required by the Director of Production Planning, the Director of Production and the Head of the planning department. These should be present at times, at least for the opening and for the result discussion. During the assessment it is considered to use a closed planning process for the evaluation. The focus of the assessment isn't the product development; the focus is the planning processes. To ensure clarity, a subdivision of processes makes the assessment area not too complex.

An experienced moderator leads the participants through the assessment. To be able to classify a processes level of maturity, a formulation of closed questions which are to be answered only with "Yes" or "No" is conducted. For this method a questionnaire was created based on the contents of the level. On the one side, the answer "Yes" doesn't automatically fulfill a criterion for a higher level plane. On the other side, the answer "No" can express the fulfillment of a criterion for a higher quality of the PPP. This effectively hinders tuning the results of the assessment by preferably answering the more positive sounding answer. Each participant evaluates or answers the questions independently and on his or her own. The answers of the questions lead to a valuation in six steps. A level is achieved, if all criteria of the actual level and all criteria of the previous level are fulfilled. Afterwards the individual valuations are presented in the whole assessment group and argued concerning the contents. The final answers are determined by the team. All team members must have answered the questions with an identical answer, at least after the discussion. If there is no consensus the question must be marked as "discordant". Disagreement means the non-fulfillment of a criterion. This consequently means it remains at a lower level. With the help of the assessment, potentials are indicated. The groundwork for a continuous improvement is thereby given. These spheres of activity are suggested by the participants of the assessment and their conversion should occur within 3-6 month in order to evolve up to the next-higher maturity level within the respective criterion. For the end of the assessment a Reporting of the result is carried out. The persons responsible decide independently whether results are worn outwardly. This entails that one of the results requests for an area as a Best-Practice and serves from now on as "a model". An internal publication of the results leads to the process partners' aiming at a constant monitoring of the consideration improvement.

5 Future prospects

The conversion and advancement of the circumscribed concept for the qualitative assessment of PPP demonstrates big potential for the applying areas. The systematic and regular disclosure of process-wise improvement potential corresponds to the CIP-approach, as it is already exercised within production. A steady increase of the efficiency in the process is the result of a continuous use of the methodology with conversion of the derived measures. By group-wide and global application of such a methodology, process experiences can be shared within several areas under circumstances noticeably more efficient. After a successful introduction of the methodology, as soon as the users are familiar with the experience of an assessment, the methodology can be extended around further consideration fields for the process maturity and suitable metrics. So further areas of action are revealed, in which continuous improvement can be monitored systematically. If the experiences and contents of the assessment have proven worthwhile, a fully automated solution is conceivable, in which the presence of a presenter is not further necessary. Besides, a derivation of the quality of the PPP at the request of the system is the future vision. In this phase a comparable base can be created around a suitable aggregation and modification of the metrics from the single areas if necessary to uncover Best Practice Processes within the group. If the assessment based on the sophisticated planning process has matured, this action of the process assessment can be transferred from general to commercial processes. A suitable adaption of the assessment contents admits a generalization of the proven action.

6 References

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